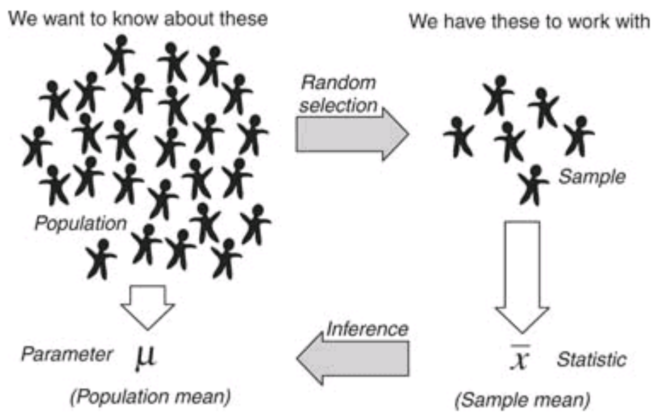
Lecture 03

Probabilistic Modeling in Hydrology-Part II (MLE & Goodness of fit tests)

ENGINEERING
COLLEGE OF ENGINEERING
SEOUL NATIONAL UNIVERSITY
서 울 대 학 교 공 과 대 학

Review on Lecture 2



❖ **Basic Probability Concepts**

- Random Variables (RV)
- Mutually Exclusive,
  Independence, Conditional Probability,
  Bayes Theorem, Total Probability Theorem, …

❖ **Types of Random Variables**

- Discrete RV & Continuous RV
- Discrete Distributions:
  Binomial, Poisson, Geometric, Negative Binomial, …
- Continuous Distributions: Uniform, Normal, Exponential, …

❖ **Chebyshev Inequality & Law of Large Numbers**

Assignment: Poisson from Binomial

Should be x

$$
\begin{aligned}
P(X=x) &= \binom{n}{k} p^x (1-p)^{n-x} \\
&= \frac{n!}{(n-x)!\,x!}\left(\frac{\lambda}{n}\right)^x \left(1-\frac{\lambda}{n}\right)^{n-x} \\
&= \frac{n(n-1)(n-2)\cdots(n-x+1)}{x!}\,\frac{\lambda^x}{n^x}\left(1-\frac{\lambda}{n}\right)^{n-x} \\
&= \frac{n(n-1)(n-2)\cdots(n-x+1)}{n^x}\,\frac{\lambda^x}{x!}\left(1-\frac{\lambda}{n}\right)^{n-x}
\end{aligned}
$$

$$\frac{n(n-1)(n-2)\cdots(n-x+1)}{n^x} \approx 1$$

$$
\begin{aligned}
\left(1-\frac{\lambda}{n}\right)^{n-x} &= \left(1-\frac{\lambda}{n}\right)^{n}\left(1-\frac{\lambda}{n}\right)^{-x} \\
&\approx e^{-\lambda}\cdot 1
\end{aligned}
$$

$$P(X=x) = \binom{n}{k} p^x (1-p)^{n-x} \approx \frac{\lambda^x e^{-\lambda}}{x!}$$

Chebyshev Inequality Example $P\{|X-\mu| \geq k\sigma\} \leq \frac{1}{k^2}$

Example 3.6. The data of table 2.1 has a mean of 66,540 cfs and a standard deviation of 22,322 cfs. Without making any distributional assumptions regarding the data, what can be said of the probability that the peak flow in a year selected at random will deviate more than 40,000 cfs from the mean?

Law of Large Numbers Example

Example 3.7. Assume that the standard deviation of peak flows on the Kentucky River near Salvisa, Kentucky, is 22,322 cfs. How many observations would be required to be at least 95% sure that the estimated mean peak flow was within 10,000 cfs of its true value if we know nothing of the distribution of peak flows?

Let $p_X(x)$ be a probability density function with mean μ_X and finite variance σ_X^2. Let $\bar{x}_n$ be the mean of a random sample of size n from $p_X(x)$. Let ε and δ be any two specified small numbers such that ($\varepsilon > 0, 0 < \delta < 1$. Then for n any integer greater than $\dfrac{\sigma_X^2}{\epsilon^2\delta}$

$$\text{prob}(|\bar{x}_n - \mu_X| \geq \epsilon) \leq \delta \qquad \textbf{(3.79)}$$

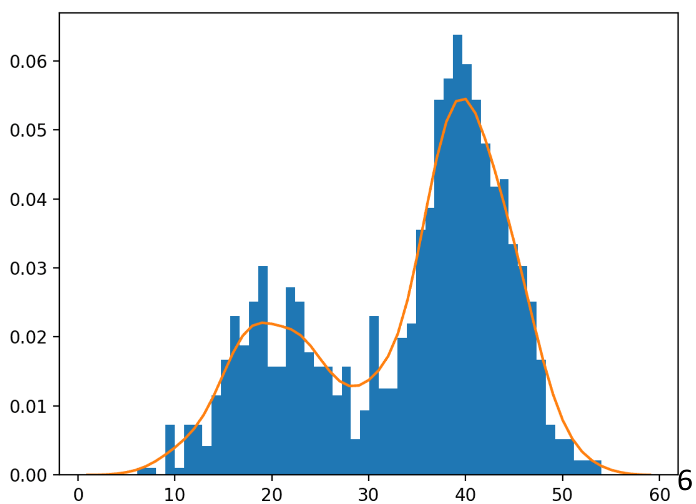
Today's Topic: Lecture 03 (Part 1)

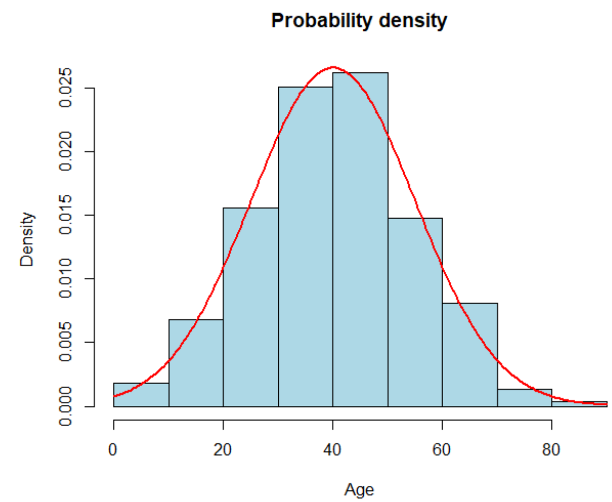
❖ Steps for Probabilistic Modeling

- ◆ Step 1: Adopt family of probability models:
  Normal, Exponential, Gamma, Binomial, Poisson
- ◆ Step 2: Fit the model to data (Estimate Parameters)

❖ Corresponding Questions to be Answered:

- ◆ Step 1: How to choose the appropriate model?
- ◆ **Step 2: How do we estimate parameters?**

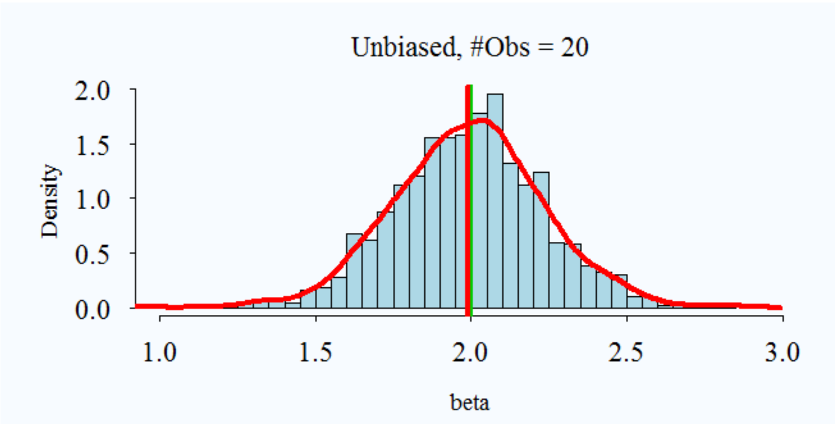


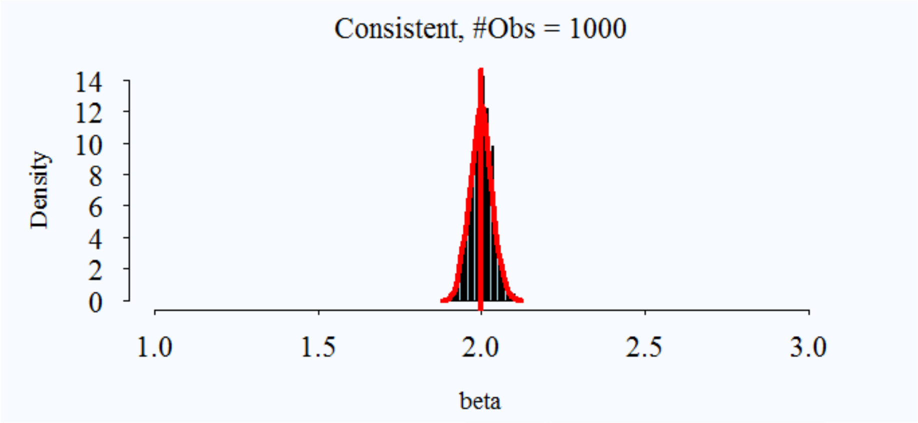


How to estimate parameters?

- Concept of Unbiasedness & Consistency
  - Bias: $E(\hat{\theta}) = \theta$
  - Consistency $\epsilon \rightarrow 0$ as $n \rightarrow \infty$ $where$ $prob(|\hat{\theta} - \theta|) = \epsilon$

- Refer to: Haan p.70-76, Handbook of Hydrology 18.1, Devore p.264-272





Unbiasedness Example

❖ Prove that the unbiased estimator or sample variance, s^2, is $\sum(x_i - \bar{x})^2/(n-1)$ using the definition of bias.

$$\begin{aligned}
\sum_{i=1}^{n}(x_i - \overline{x})^2 &= \sum_{i=1}^{n}\left(x_i^2 - 2x_i\overline{x} + \overline{x}^2\right) \\
&= \sum_{i=1}^{n} x_i^2 - 2\overline{x}\sum_{i=1}^{n} x_i + \sum_{i=1}^{n}\overline{x}^2 \\
&= \sum_{i=1}^{n} x_i^2 - 2n\overline{x}^2 + n\overline{x}^2 \\
&= \sum_{i=1}^{n} x_i^2 - n\overline{x}^2
\end{aligned}$$

Maximum Likelihood Estimation (MLE)

❖ Framework behind MLE

- Assume there is a true, known data generation process X from $f(x|\theta)$, with true but unknown parameter θ
- Estimator ($\hat{\theta}$) is a function of RVs (X, x_i)
- Estimate is a particular value of $\hat{\theta}$ for observed data x

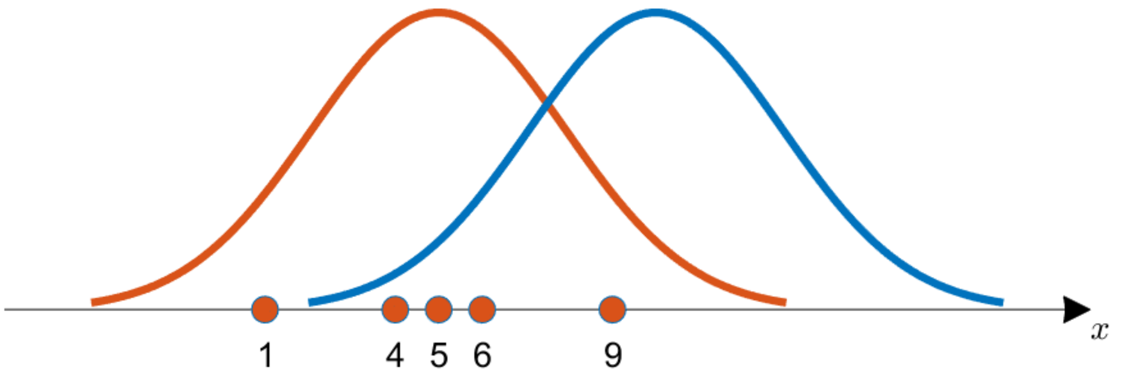
❖ Procedures for MLE

- Pick θ to make observations likely under $f(x|\theta)$
- Likelihood function $\mathrm{L}(\theta)$

$$\mathrm{L}(\theta) = \prod_{i=1}^{n} f(x_i|\theta)$$

- Choose θ to maximize $\mathrm{L}(\theta)$ => $\hat{\theta}_{MLE}$

Concept of MLE



MLE Calculation Example (1): Normal

MLE Calculation Example (2): Exponential

Sampling Variability in MLE

❖ Sampling Variability

- $\hat{\lambda} = 0.136$ is an estimate of true (unknown) λ
- $\hat{\lambda}$ will vary if based on a different sample of data
  - Less data => More Sampling variability
  - More data => More Sampling => More like Normal Distribution

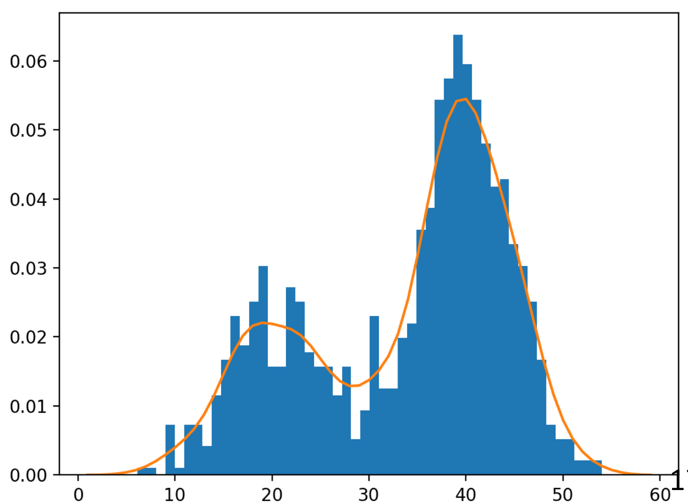
❖ Approximate Normality of MLE

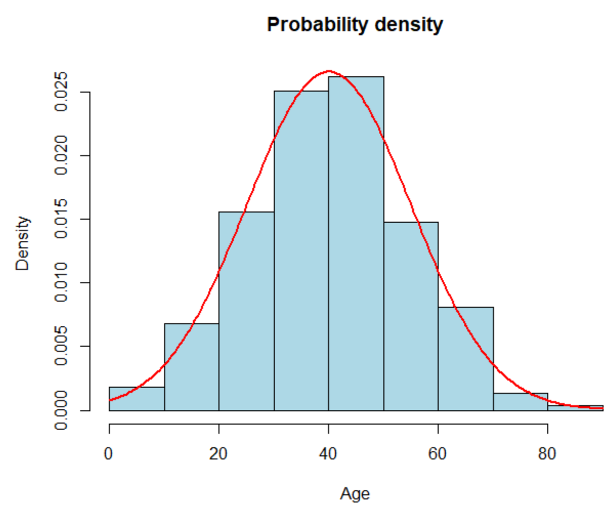
- For 1 parameter models (exponential distribution), as n gets big,

$$\hat{\theta}_{MLE} \sim N(\theta_0, I(\theta_0)^{-1})$$

Today's Topic: Lecture 03 (Part 2)

- ❖ Steps for Probabilistic Modeling
  - ◆ Step 1: Adopt family of probability models:
    Normal, Exponential, Gamma, Binomial, Poisson
  - ◆ Step 2: Fit the model to data (Estimate Parameters)
- ❖ Corresponding Questions to be Answered:
  - ◆ **Step 1: How to choose the appropriate model?**
  - ◆ Step 2: How do we estimate parameters?





How to pick a model? (1)

❖ 어떤 데이터를 가지고 있을 때, 알맞는 분포는 어떻게 결정하는가?

[1] Physical Basis (물리적인 특성을 바탕으로 선택)

◆ 예를 들어, 아래 성질을 가진 데이터는 _________ 분포를 따른다.

- Events are Independent
- Rate of occurrence is constant
- Number of extreme events per year (in hydrology)

How to pick a model? (2)

❖ 어떤 데이터를 가지고 있을 때, 알맞는 분포는 어떻게 결정하는가?

[2] Limit Laws (극한 정리)

◆ Theory suggest a distribution as $\boldsymbol{n} \rightarrow \infty$

◆ For instance, let X = daily precipitation and $X \sim exp(\lambda)$

◆ How about $\bar{X} = \frac{1}{365} \sum X_i$? (Annual average precipitation?)

How to pick a model? (3)

❖ 어떤 데이터를 가지고 있을 때, 알맞는 분포는 어떻게 결정하는가?

[3] Empirical Basis: Which model fits data best?

◆ **Goodness-of-fit Approaches** such as:

- Picking based on Scores (AIC)
- Visual Assessments (P-P or Q-Q Plots)
- Hypothesis testing (Chi-squared test)

How to pick a model? (3)

❖ AIC (Akaikes Information Criterion, 아카이케 정보 기준)

$$AIC = -2\times l\left(\hat{\theta}_{MLE}\right) + 2\times(\#\ of\ par.)$$

- Goal: pick the model with smallest AIC score
  (Large maximum likelihood, penalty for model parameters)
- Rule of thumb

How to pick a model? (3)

❖ AIC Example

How to pick a model? (3)

❖ Probability plots and Quantile plots (P-P or Q-Q plots)

- ◆ Testing the plausibility that the observed data $x_1, \ldots, x_n$ is drawn from a distribution with CDF F(X)
- ◆ Strategy: for proposed distribution and data, plot evaluated under model against empirical (model free) estimates
- ◆ Procedures

P-P & Q-Q Plot Examples

How to pick a model? (3)

- Goodness of Fit Tests (Chi-squared)
  - Define null hypothesis H_0
  - Define alternative hypothesis H_A
  - Identify a test statistics
  - Determine the null distribution: sampling distribution of test statistic under H_0
  - Compare test statistics to null distribution

How to pick a model? (3): Example

Example 5.28. Goodness-of-fit of wet runs of rainfall. In Example 4.16 we considered the distributions of wet and dry runs of daily rainfall at Kew in London, England, from January 1958 to May 1965. It was seen in Table 4.1.5 and Fig. 4.1.12 that the geometric distribution provides a close fit to the observed distribution of wet runs. For a formal assessment of these fits, we can use the chi-squared goodness-of-fit criterion, as approximated by Eq. (5.6.1), using the given data.

Null hypothesis H_0: The random variable (wet runs) has a geometric distribution with $p = 0.392$.

Alternate hypothesis H_1: The random variable does not have the specified distribution.

Level of significance: $\alpha = 0.05$.

Critical region: $X^2 \geq \chi^2_{8,0.05} = 15.5$ (from Table C.3), noting that in Table 4.1.5 there are $l = 10$ classes and $k = 1$ for the application of Eq. (5.6.1), which gives $\nu = 10 - 1 - 1 = 8$.

Calculations: From Table 4.1.5,

$$X^2 = \frac{(194-179.6)^2}{179.6} + \frac{(101-109.2)^2}{109.2} + \frac{(66-66.4)^2}{66.4} + \frac{(30-40.3)^2}{40.3} + \frac{(26-24.5)^2}{24.5} + \frac{(11-14.9)^2}{14.9} + \frac{(13-9.1)^2}{9.1} + \frac{(7-5.5)^2}{5.5} + \frac{(5-3.3)^2}{3.3} + \frac{(2-2.2)^2}{2.2} = 8.49.$$

Decision: Because $X^2 < \chi^2_{8,0.05} = 15.5$, the null hypothesis is not rejected. It can be concluded that the wet runs have a geometric distribution as specified.

In a practical sense we should be aware, however, that the true distribution of wet runs cannot be exactly the same as the hypothesized distribution. What we have found is a good approximation to the true distribution.

Regarding the dry runs, the logarithmic series distribution seems to provide a better fit, as seen from Table 4.1.6 and Fig. 4.1.13. The chi-squared test can also be applied to this hypothesis.

Table 4.1.5 Distributions of wet runs[a]

Length of wet run in days, (1)	Observed distribution, (2)	Product of (1) × (2), (3)	Geometric distribution, (4)
1	194	194	179.6
2	101	202	109.2
3	66	198	66.4
4	30	120	40.3
5	26	130	24.5
6	11	66	14.9
7	13	91	9.1
8	7	56	5.5
9	5	45	3.3
10	2	20	2.2
11 + (15.3)	3	46	
Totals	458	1168	

Observed from January 1958 to May 1965 at Kew, London, England.
[a] *Note*: 15.3 represents mean length of 3 runs longer than 10.
Wet run: a period of consecutive rainy days for which the day immediately before the p
immediately after the period are dry.
Source: Data from Chatfied (1966); obtained with the permission of the publishers.

ENGINEERING
COLLEGE OF ENGINEERING
SEOUL NATIONAL UNIVERSITY
서 울 대 학 교 공 과 대 학

복습과제 (Quiz 1, Lec 01-Lec 03)

- ❖ 책
- ❖ 문제

Future Schedule Preview

❖ 다음 주 강의 (2022-09-27 화요일 18:30-21:20)

◆ ***Training with Probabilistic Modeling***

- **Bring LAPTOPS with R & R studio on Sep-27!**
- Quiz 1 before the Training Session * Bring Calculators
  Multiple Questions, Short Answers, Problem Solving (~ 5-6 Q.)

❖ Other Assignments (Plans for September)

◆ Datacamp Assignments due 9/15 (Introduction to R)

- Introduction to R due 9/15
- Foundations of Probability in R due 9/23

◆ Training Assignment (on 9/27) due 10/04